Pocket SDR - An Open-Source GNSS SDR, ver. 0.15b

Overview

Pocket SDR is an open-source Global Navigation Satellite System (GNSS) receiver based on software-defined radio (SDR) technology. It consists of RF frontend devices named **Pocket SDR FE**, utilities for these devices, and GNSS SDR applications (APs) written in Python, C, and C++. It supports almost all signals for **GPS**, **GLONASS**, **Galileo**, **QZSS**, **BeiDou**, **NavIC**, and **SBAS**.

The Pocket SDR FE device includes 2, 4, or 8 RF frontend channels, supporting the GNSS L1 band (1525 - 1610 MHz) or L2/L5/L6 bands (1160 - 1290 MHz). For these signal bands, refer to **GNSS Signal Bands**. Each RF channel of the Pocket SDR FE device provides IF (intermediate-frequency) bandwidth of up to 36 MHz. The ADC sampling rate can be configured up to 32 Msps (FE 2CH) or 48 Msps (FE 4CH and FE 8CH).

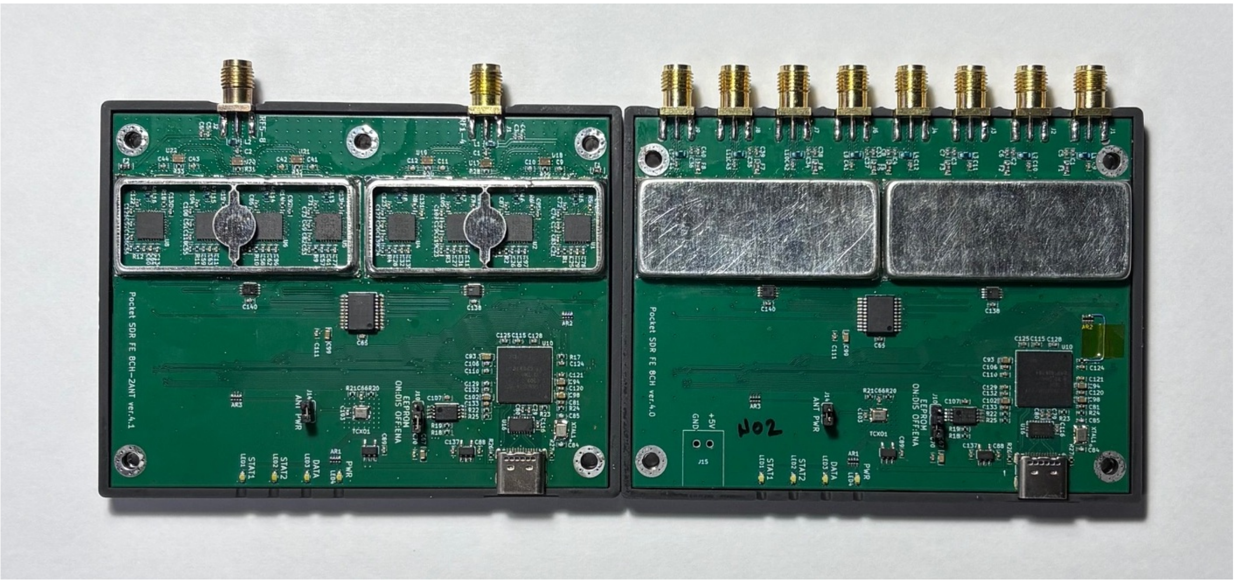
Pocket SDR also includes utility programs to configure the Pocket SDR FE devices, capture, and dump digitized IF data. Additionally, Pocket SDR provides GNSS SDR APs to display the PSD (power spectrum density) of captured IF data, search for GNSS signals, track these signals, decode navigation data, and generate PVT (position, velocity, and time) solutions. These utilities and APs are compatible with **Windows**, **Linux**, **Raspberry Pi OS**, **macOS**, and other environments.

The supported GNSS signals by Pocket SDR are as follows. For details on these signals and Pocket SDR signal IDs, refer to **GNSS Signals**.

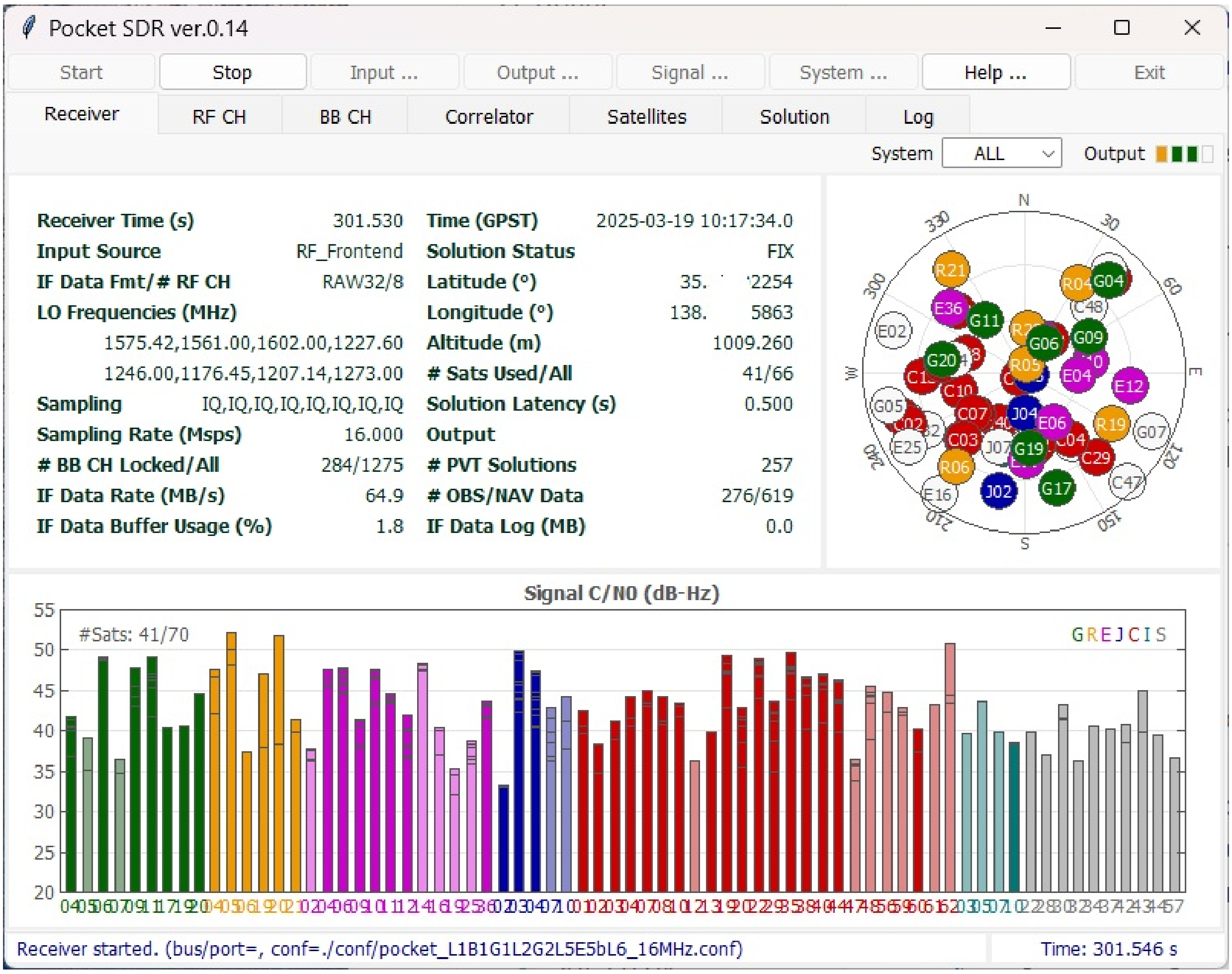
- **GPS**: L1C/A, L1C-D, L1C-P, L2C-M, L5-I, L5-Q
- **GLONASS**: L1C/A (L1OF), L2C/A (L2OF), L1OCd, L1OCp, L2OCp, L3OCd, L3OCp
- **Galileo**: E1-B, E1-C, E5a-I, E5a-Q, E5b-I, E5b-Q, E5AltBOC (E5abQ), E6-B, E6-C
- **QZSS**: L1C/A, L1C/B, L1C-D, L1C-P, L1S, L2C-M, L5-I, L5-Q, L5S-I, L5S-Q, L6D, L6E
- **BeiDou**: B1I, B1C-D, B1C-P, B2a-D, B2a-P, B2I, B2b-I, B3I
- **NavIC**: L1-SPS-D, L1-SPS-P, L5-SPS
- **SBAS**: L1C/A, L5-I, L5-Q

These utilities and APs are written in Python, C, and C++ in a compact and modular way, making them easy to modify for adding custom algorithms.

Pocket SDR FE 2CH (v.2.0 and v.2.3), FE 4CH



Pocket SDR FE 8CH (2-RF-inputs and 8-RF-inputs)



GUI-based Real-Time GNSS SDR Receiver AP

The introduction to Pocket SDR is available in the following slides:

T.Takasu, An Open Source GNSS SDR: Development and Application, IPNTJ Next GNSS Technology WG, February 21, 2022 (https://gpspp.sakura.ne.jp/paper2005/IPNTJ_NEXTWG_202202.pdf)

For an application of Pocket SDR, refer to the following slides:

T.Takasu, Development of QZSS L6 Receiver without Pilot Signal by using SDR, IPNTJ Annual Conference, June 10, 2022 (https://gpspp.sakura.ne.jp/paper2005/IPNTJ_20220610.pdf)

For design and implementation details, refer to the following slides:

T.Takasu, Pocket SDR: Design, Implementation and Applications, A seminar for GNSS Software Defined Receivers, Nov 19, 2024 (https://gpspp.sakura.ne.jp/paper2005/pocketsdr_seminar_202411_revA.pdf)

Directory Structure and Contents

```
PocketSDR
├── bin          # Pocket SDR APs binary programs
├── app          # Pocket SDR APs source programs
│   ├── pocket_conf  # Pocket SDR FE device configurator
│   ├── pocket_dump  # Dump digital IF data of Pocket SDR FE device
│   ├── pocket_scan  # Scan and list USB devices
│   ├── pocket_acq   # GNSS signal acquisition
│   ├── pocket_trk   # GNSS signal tracking and PVT generation
│   ├── pocket_snap  # Snapshot Positioning
│   ├── pocket_calib # Antenna array attitude / per-CH bias calibration
│   ├── convbin      # RINEX converter supporting Pocket SDR
│   └── str2str      # Stream converter (RTKLIB-derived)
├── src          # Pocket SDR library source programs
├── python       # Pocket SDR Python scripts (incl. pocket_sdr.py GUI)
├── lib          # Libraries for APs and Python scripts
│   ├── win32        # Built libraries for Windows (UCRT64)
│   ├── macos        # Built libraries for macOS
│   ├── linux        # Built libraries for Linux or Raspberry Pi OS
│   ├── win32_msvc   # Built libraries for Windows (MSVC, optional)
│   ├── build        # Makefiles to build libraries (UCRT64 / Linux /
macOS)
│   ├── build_msvc   # Makefiles to build libraries with MSVC (optional)
│   ├── cyusb        # Cypress EZ-USB API (CyAPI.a) and includes
│   ├── RTKLIB       # RTKLIB source programs based on 2.4.3 b34
│   ├── pocketfft    # PocketFFT (header-only FFT, used internally)
│   ├── SoapySDR     # SoapySDR headers + import lib for SDR device support
│   ├── (libfec)     # Library for FEC (forward error corrections) ([1])
│   └── (LDPC-codes) # Library for LDPC-decoder ([2])
├── conf         # Configuration files for Pocket SDR FE
├── FE_2CH       # Pocket SDR FE 2CH H/W and F/W
├── FE_4CH       # Pocket SDR FE 4CH H/W and F/W
├── FE_8CH       # Pocket SDR FE 8CH H/W and F/W (3-bit RF samples)
├── FE_8CH_12bit # Pocket SDR FE 8CH 12-bit H/W and F/W (FPGA-based, in
dev.)
├── driver       # Driver installation instruction for Pocket SDR FE
├── doc          # Documents (incl. api_ref.md, command_ref.md)
├── image        # Image files for documents
├── sample       # Sample digital IF data captured by Pocket SDR FE
└── test         # Test codes and data

Note: Items in parentheses () are not included in the package and are
fetched by lib/clone_lib.sh.
```

Installation for Windows

- Download and extract PocketSDR.zip or clone the git repository **here** to an appropriate directory <install_dir>. to an appropriate directory <install_dir>.

```
> unzip PocketSDR.zip
or
> git clone https://github.com/tomojitakasu/PocketSDR
```

- Install USB device driver for Pocket SDR FE according to **driver/readme.txt** .
- Install **Python** with checking "Add python.exe to PATH".
- Install additional python packages as follows.

```
> pip install numpy scipy matplotlib
```

- (Optional) To use SoapySDR-supported devices (LimeSDR, HackRF, RTL-SDR, etc.) with the python APs, install **radioconda**. radioconda is the recommended way to get SoapySDR + per-device plug-in DLLs on Windows in one bundle. Steps:
    1. Download the latest `Radioconda-Windows-x86_64.exe` from the releases page and run it with default settings (per-user install, no admin needed).
    2. Add the runtime DLL directory to the Windows PATH so `SoapySDR.dll` and the per-device plug-in DLLs can be loaded:

```
<radioconda_install_dir>\Library\bin
```

```
 (typical default: `C:\Users\<user>\radioconda\Library\bin`)
```

3. Verify the SoapySDR install and the plug-in modules in a fresh command prompt:

```
> SoapySDRUtil --info
> SoapySDRUtil --probe="driver=lime"
```

```
 The first command lists the discovered modules
 (e.g., `LMS7`, `HackRF`, `RTLSDR`, `SDRPlay`, `BladeRF`).
 The second probes a specific device. If the module is missing, see
 radioconda's docs to add per-device packages via `mamba install`.
```

4. The Pocket SDR python APs (`pocket_sdr.py`, `pocket_dump.py`, ...) auto-add radioconda's `Library\bin` to the DLL search path on start, so PATH addition above is mainly needed for the C-level `pocket_trk.exe` etc. to find the DLLs at runtime.

- Add the Pocket SDR binary programs path (<install_dir>\PocketSDR\bin) to the command search path (Path) of Windows environment variables.
- Add the Pocket SDR Python scripts path (<install_dir>\PocketSDR\python) to the command search path (Path) of Windows environment variables.

To rebuild binary programs for Windows

- If you want to rebuild the binary utilities, APs, or shared libraries for the python APs for Windows, you need **MSYS2** with the **UCRT64** environment. The UCRT64 toolchain links against the same C runtime (`ucrtbase.dll`) as **radioconda**'s SoapySDR DLLs, which gives stable high-rate streaming with SDR hardware (e.g., LimeSDR at 64 Msps). The legacy **MINGW64** environment also builds, but suffers significant performance loss with SoapySDR devices due to msvcrt/UCRT mixing.
- Open the **MSYS2 UCRT64** shell and install the development tools and the Python stack used by the python APs:

```
$ pacman -Syy
$ pacman -S git make
$ pacman -S mingw-w64-ucrt-x86_64-gcc
$ pacman -S mingw-w64-ucrt-x86_64-binutils
$ pacman -S mingw-w64-ucrt-x86_64-python
$ pacman -S mingw-w64-ucrt-x86_64-python-numpy
$ pacman -S mingw-w64-ucrt-x86_64-python-scipy
$ pacman -S mingw-w64-ucrt-x86_64-python-matplotlib
```

- If you intend to use SoapySDR-supported devices (LimeSDR, HackRF, RTL-SDR, SDRPlay, BladeRF, etc.), install **radioconda** (see Windows install section above for details). The Pocket SDR build picks up the SoapySDR headers and the import library from the radioconda install:
    - Headers: `<radioconda>\Library\include\SoapySDR\`
    - DLL: `<radioconda>\Library\bin\SoapySDR.dll`
    - The MinGW import library `libSoapySDR.dll.a` is auto-generated from `SoapySDR.dll` during the first library build via `gen_soapysdr_implib.sh`.

  The default radioconda location used by the Makefiles is `/c/Users/<user>/radioconda/Library`. If installed elsewhere, override with `SOAPY_ROOT` on the make command line (see below).
- Move to the library directory. The external libraries are for Forward Error Correction (FEC) and Low-Density Parity-Check (LDPC) decoding. Install the external library source trees ([1], [2]) as follows:

```
$ cd <install_dir>/lib
$ ./clone_lib.sh
```

- Move to the library build directory and build libraries.

```
$ cd <install_dir>/lib/build
$ make
$ make install
```

The default radioconda location is `/c/Users/<user>/radioconda/Library`. If installed elsewhere, override on the command line:

```
$ make SOAPY_ROOT=/c/path/to/radioconda/Library
```

- Move to the application program directory and build utilities and APs.

```
$ cd <install_dir>/app
$ make
$ make install
```

Installation for Linux or Raspberry Pi OS

> **Note**: Pocket SDR has been primarily verified on Ubuntu / Debian-based Linux distributions. **Raspberry Pi OS** is supported in principle but SoapySDR-based device support has not been thoroughly verified on Raspberry Pi.

- You need fundamental development packages and some libraries. Confirm the following packages installed: git, gcc, g++, make, libusb-1.0-0-dev, libfftw3-dev, python3, python3-numpy, python3-scipy, python3-matplotlib, python3-tk
- (Optional) To use SoapySDR-supported devices (LimeSDR, HackRF, RTL-SDR, SDRPlay, BladeRF, etc.), install SoapySDR and the per-device modules. On Ubuntu / Debian, the simplest way is the meta-package `soapysdr-module-all`, which pulls in all device modules at once:

```
$ sudo apt install libsoapysdr-dev soapysdr-tools soapysdr-module-all
```

If you prefer to install only the modules you need, install them individually instead:

```
$ sudo apt install libsoapysdr-dev soapysdr-tools
$ sudo apt install soapysdr-module-rtlsdr      # RTL-SDR
$ sudo apt install soapysdr-module-hackrf      # HackRF
$ sudo apt install soapysdr-module-lms7        # LimeSDR (also: limesuite)
$ sudo apt install soapysdr-module-bladerf     # BladeRF
$ sudo apt install soapysdr-module-sdrplay     # SDRPlay (non-free, see vendor)
$ sudo apt install soapysdr-module-airspy      # Airspy
```

After install, verify the modules are loadable:

```
$ SoapySDRUtil --info
$ SoapySDRUtil --probe="driver=lime"   # for LimeSDR
```

Some devices (LimeSDR / SDRPlay / etc.) need additional vendor drivers and udev rules — refer to each project's documentation.

For distros without prebuilt packages, build SoapySDR and the device modules from source: see https://github.com/pothosware/SoapySDR/wiki .

- Download and extract PocketSDR.zip or clone the git repository to an appropriate directory <install_dir>.

```
$ unzip PocketSDR.zip
or
$ git clone https://github.com/tomojitakasu/PocketSDR
```

- Move to the library directory and install the external library source trees ([1], [2]) as follows:

```
$ cd <install_dir>/lib
$ chmod +x clone_lib.sh
$ ./clone_lib.sh
```

- Move to the library build directory and build libraries.

```
$ cd <install_dir>/lib/build
$ make
$ make install
```

- Move to the application program directory and build utilities and APs.

```
$ cd <install_dir>/app
$ make
$ make install
```

- Add the Pocket SDR binary programs path (<install_dir>/PocketSDR/bin) to the command search path.
- For Linux or Raspberry Pi OS users, you usually need to have root permission to access USB devices. So you have to add "sudo" to execute some utilities, APs or python scripts like:

```
$ sudo pocket_conf ../conf/pocket_L1L6_12MHz.conf
$ sudo pocket_dump -t 10 ch1.bin ch2.bin
$ sudo pocket_sdr.py
```

Installation for macOS

> **Note**: macOS support is provided on a best-effort basis. Build and basic functionality have been confirmed but **SoapySDR-based device support on macOS is not actively verified**. SoapySDR may be installed via Homebrew (`brew install soapysdr` plus per-device formulas such as `soapyhackrf`, `soapyrtlsdr`, `limesuite`) but expect to do some integration work yourself.

- You need **Homebrew** as a package manager for macOS. Install Homebrew by running the following command in the terminal:
- Open a terminal window on macOS and install the basic libraries by using Homebrew:

```
$ brew install numpy
$ brew install scipy
$ brew install python3-matplotlib
$ brew install python3-numpy
$ brew install python3-tk
$ brew install libusb
```

- Download and extract PocketSDR.zip or clone the git repository to an appropriate directory <install_dir>.

```
$ unzip PocketSDR.zip
or
$ git clone https://github.com/tomojitakasu/PocketSDR
```

- Move to the library directory, install external library source trees ([1], [2]) as follows:

```
$ cd <install_dir>/lib
$ chmod +x clone_lib.sh
$ ./clone_lib.sh
```

- Move to the library build directory and build libraries.

```
$ cd <install_dir>/lib/build
$ make
$ make install
```

- Move to the application program directory and build utilities and APs.

```
$ cd <install_dir>/app
$ make
$ make install
```

- Add the Pocket SDR binary programs path (<install_dir>/PocketSDR/bin) to the command search path.

GNSS SDR Utilities and APs

Pocket SDR includes the following utilities for the Pocket SDR FE.

- **pocket_scan**: Scans and lists USB Devices.
- **pocket_conf**: Configures Pocket SDR FE device.

- **pocket_dump**: Captures and dumps IF data from the Pocket SDR FE device

Pocket SDR also provides the following GNSS SDR APs:

- **pocket_psd.py** : Plots PSD and histograms of IF data.
- **pocket_acq.py** : Performs GNSS signal acquisition from IF data.
- **pocket_trk.py** : Tracks GNSS signals and decodes navigation data in IF data.
- **pocket_snap.py**: Executes snapshot positioning using captured IF data.
- **pocket_sdr.py** : A GUI-based GNSS SDR receiver application.
- **pocket_plot.py**: Plots receiver logs generated by pocket_trk or pocket_sdr.py.
- **pocket_acq** : A C-version of pocket_acq.py (w/o graphical plots).
- **pocket_trk** : A C-version of pocket_trk.py (w/o graphical plots).
- **pocket_snap** : A C-version of pocket_snap.py.

For more details about these utilities and APs, please refer to **Pocket SDR Command References**.

GUI-based Real-Time GNSS SDR Receiver AP

Starting from version 0.13, Pocket SDR includes a GUI-based real-time GNSS SDR receiver AP, **pocket_sdr.py**. To execute the application, follow these steps:

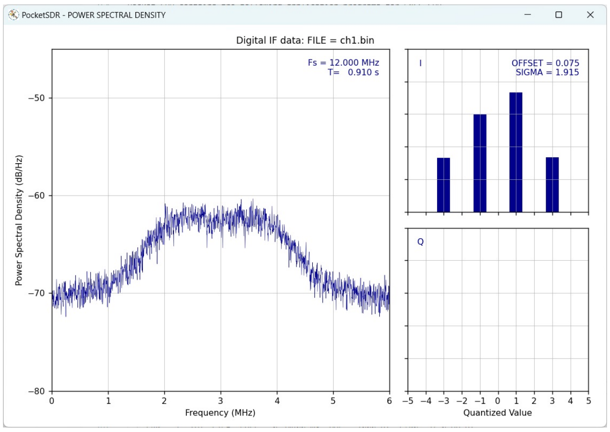
```
$ chmod +x <install_dir>/python/pocket_sdr.py
$ sudo ./<install_dir>/python/pocket_sdr.py
or alternatively:
$ sudo python <install_dir>/python/pocket_sdr.py
```

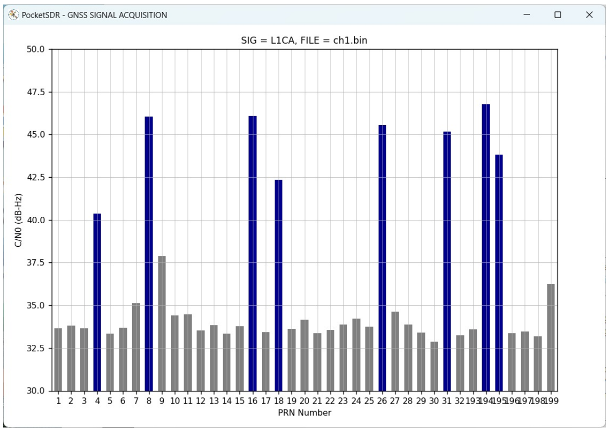
For more information about this application, please refer to **pocket_sdr.py help**.

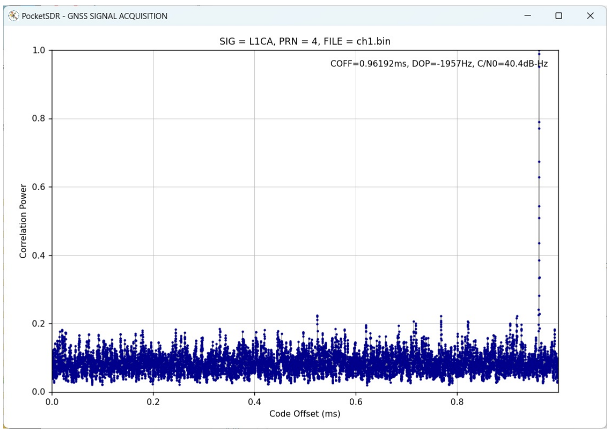
Execution Examples of Utilities and APs

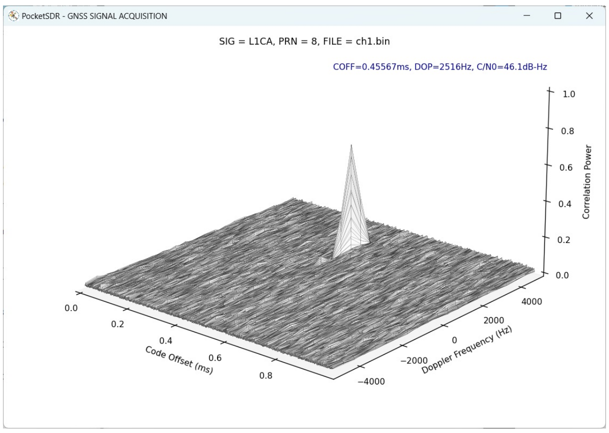
Here are some examples of how to execute the utilities and the APs:

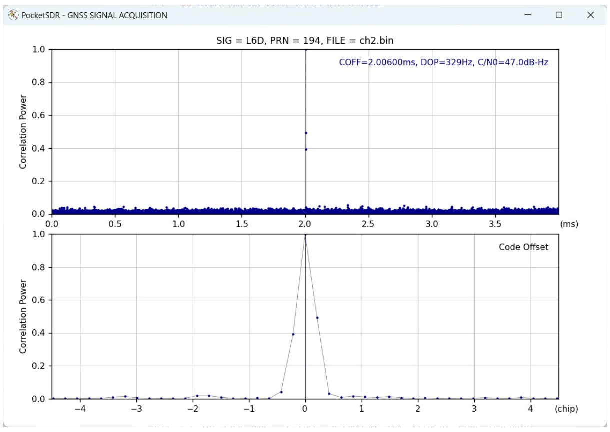
```
$ sudo pocket_conf
...
$ sudo pocket_conf conf/pocket_L1L6_12MHz.conf
Pocket SDR device settings are changed.

$ sudo pocket_dump -t 5 ch1.bin ch2.bin
  TIME(s)    T   CH1(Bytes)   T   CH2(Bytes)   RATE(Ks/s)
      5.0    I     60047360  IQ    120094720      11985.5

$ pocket_psd.py ch1.bin -f 12 -h

$ pocket_acq.py ch1.bin -f 12 -sig L1CA -prn 1-32,193-199
SIG= L1CA, PRN=   1, COFF=  0.23492 ms, DOP= -1519 Hz, C/N0= 33.6 dB-Hz
SIG= L1CA, PRN=   2, COFF=  0.98558 ms, DOP=  2528 Hz, C/N0= 33.8 dB-Hz
SIG= L1CA, PRN=   3, COFF=  0.96792 ms, DOP=  3901 Hz, C/N0= 33.7 dB-Hz
SIG= L1CA, PRN=   4, COFF=  0.96192 ms, DOP= -1957 Hz, C/N0= 40.4 dB-Hz
...
$ pocket_acq.py ch1.bin -f 12 -sig L1CA -prn 4
...
$ pocket_acq.py ch1.bin -f 12 -sig L1CA -prn 8 -3d
...
$ pocket_acq.py ch2.bin -f 12 -sig L6D -prn 194 -p
...
$ pocket_trk.py ch1.bin -f 12 -sig L1CA -prn 1-32
...
$ pocket_trk.py ch1.bin -f 12 -sig E1B -prn 18 -p
...
$ pocket_trk.py ch2.bin -f 12 -sig E6B -prn 4 -log trk.log -p -ts 0.2
...
```

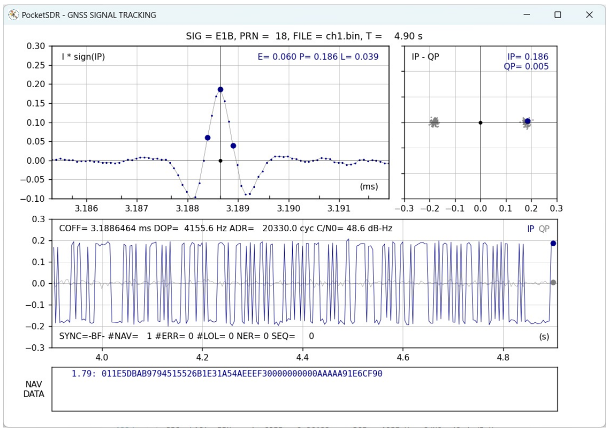


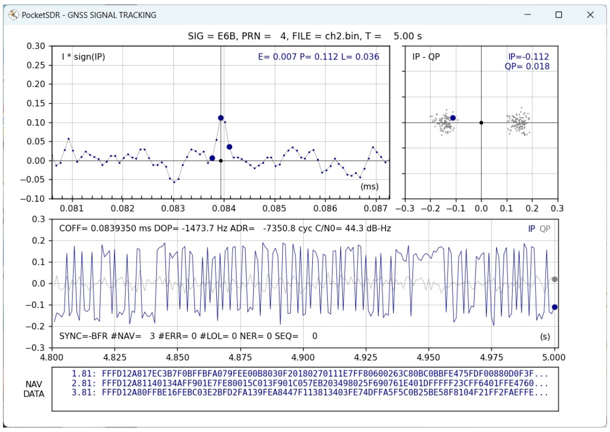












References

[1] **libfec**: A library for forward error correction (FEC).

[2] **LDPC-codes**: A library for LDPC (Low-Density Parity-Check) decoding.

[3] *NMEA 0183: Standard for Interfacing Marine Electronic Devices*, National Marine Electronics Association and International Marine Electronics Association, 2013.

[4] *RTCM 10403.4 with Amendment 1: Differential GNSS (Global Navigation Satellite Systems) Service - Version 3*, Radio Technical Commission for Maritime Services, November 1, 2024

History

- **2021-10-20 (v0.1)**: Initial draft version.
- **2021-10-25 (v0.2)**: Added rebuild firmware and write firmware image to Pocket SDR.
- **2021-12-01 (v0.3)**: Added and modified Python scripts.
- **2021-12-25 (v0.4)**: Added and modified Python scripts.
- **2022-01-05 (v0.5)**: Fixed several issues.
- **2022-01-13 (v0.6)**: Added and modified Python scripts.
- **2022-02-15 (v0.7)**: Improved performance and added more Python scripts.
- **2022-07-08 (v0.8)**: Added C-version of pocket_acq.py and pocket_trk.py.
- **2024-01-03 (v0.9)**: Added C-version of pocket_snap.py. pocket_trk now supports multi-signal and multi-threading.

- **2024-01-12 (v0.10)**: Added support for NavIC L1-SPS-D, L1-SPS-P, GLONASS L1OCd, L1OCp, and L2OCp.
- **2024-01-25 (v0.11)**: Added support for decoding GLONASS L1OCd NAV data and NB-LDPC error correction for BDS B1C, B2a, and B2b.
- **2024-05-28 (v0.12)**: Performance optimizations. Added support for PVT generation, RTCM3, and NMEA outputs.
- **2024-07-04 (v0.13)**: Added GUI-based GNSS SDR receiver AP. Added support for macOS and Raspberry Pi OS.
- **2025-03-21 (v0.14)**: Added Pocket SDR FE 8CH. Fixed various issues.
- **2026-05-02 (v0.15b)**: Added antenna array calibration (per-epoch EKF on single-difference carrier-phase residuals) and digital beam-forming. Added LUT-based array combine (~2x throughput at narch=8). Added GUI Array tab with calibration / beam control and gain pattern overlay. Added Galileo E5ABQ signal support. Added SoapySDR / LimeSDR support on Windows (UCRT64 + radioconda). Refactored array calibration / beam-forming API.
